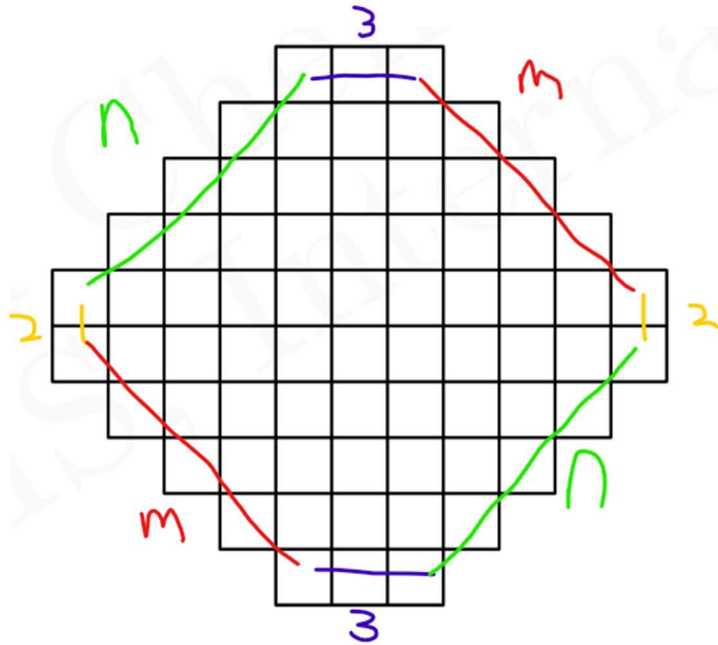


2 hours on examples

Problem 3 (NIMO)

We denote $a_{n,m}$ to be the number of ways to fill such an “special octagonal figure” where the top and bottom edges have three squares each, the left and right edges have two squares each, the left-top and bottom-right edges have n squares each, and the remaining edges have m squares each.

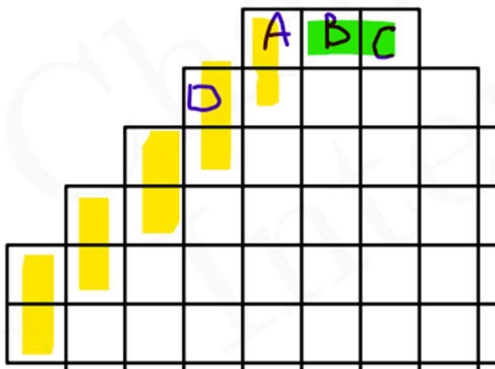


For example, the diagram portrayed would be $a_{3,3}$.

Claim 1: $a_{n,m} = a_{n-1,m-1} + a_{n-1,m} + a_{n,m-1}$ for $n, m > 1$

Label the top three squares A, B, C from left to right.

Case 1: A is filled with a vertical domino, B is filled with a horizontal domino.



Because A is filled with a vertical domino, note that the cell (labeled D) diagonally adjacent to A to the left-below must be filled with a vertical domino as well. This continues to the left edge.

We then have B filled with a horizontal domino, which must cover C as well. There are then $a_{n,m-1}$ ways to fill the remaining grid.

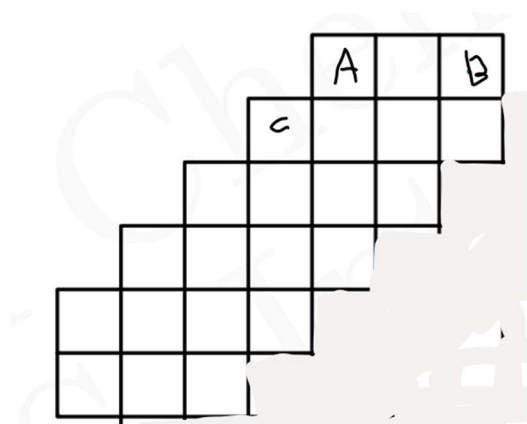
Case 2: A is filled with a vertical domino, B is filled with a vertical domino.

There must be a vertical domino filling D and so on once again. Note that a similar thing occurs on the other side as well, with C must being filled with a vertical domino. The top two cells in each column must be covered by the same domino. This results in $a_{n-1,m-1}$ cases.

Case 3: A is filled with a horizontal domino.

C must then be filled with a vertical domino. Following analogous steps, we'd get $a_{n-1,m}$ ways.

Claim 2: $a_{n,0} = a_{n-1,0} + 2$ for $n > 1$, $a_{0,m} = a_{0,m-1} + 2$ for $m > 1$.

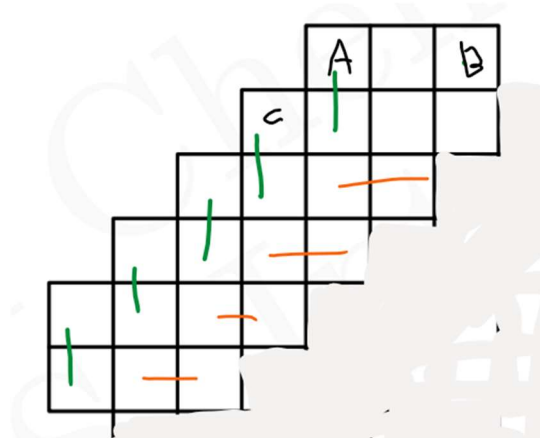


Case 1: We fill the cell labeled A horizontally.

The cell labeled B must be filled vertically, giving us $a_{n-1,0}$.

Case 2: We fill the cell labeled A vertically.

We would then have to fill C vertically and so on (green). Eventually, there is a corner in the bottom-left and we must fill in the orange spaces. The remaining 2×2 may be filled in two ways.



Note that $a_{0,0}$ is a 2×3 rectangle which, case-working, may be filled in three ways. Therefore,

$$a_{1,0} = a_{0,1} = 5; a_{2,0} = a_{0,2} = 7 \dots$$

We then have, doing the calculation, $a_{5,5} = 1683$.

Problem 5 (15USATST5)

We claim that the answer is $\lfloor \log_2 n \rfloor$. We equivalently prove that, with x colors, we may have at most 2^x vertices. We show the bound and attainability with induction.

We start with attainability.

Base Case: $n = 1$. Trivially, the answer is zero.

Inductive Step: $n = \{2^x + 1, \dots, 2^{x+1}\} \rightarrow \{2^{x+1} + 1, \dots, 2^{x+2}\}$

For $n \geq 2$, we may write $n = 2^x + m$ for some $1 \leq m \leq 2^x$. We prove that we need at least $x + 1$ colors. We may split the n vertices into two groups A, B of sizes $2^x, m$ respectively.

To prove attainability, we first orient and label the edges that run within A and B following the induction hypothesis. For all the edges running between A and B , point the edge from the vertex in group A to the vertex in group B and color it with the different color. Therefore, a pair of two consecutive directed edges may not traverse between groups A, B twice, pairs that traverse between the groups once would involve one edge of a distinct color, and pairs that do not traverse between the groups would work by the inductive hypothesis.

We now go on to showing the bound. In such a tournament, we fix a color and look at all the edges of that color. Each vertex in our grid is either the start of one of these edges, the end of one of these edges, or neither. We let A be the set of vertices that are the start of one of the directed edges and B be all the other vertices. From $|A| \leq \frac{n}{2}$ we have $|B| \geq \frac{n}{2}$. Therefore, it takes at least x colors apart from the one removed to color B by the inductive hypothesis.

Conclusion: By induction $\lfloor \log_2 n \rfloor$ is the answer.

8 clubs; 4.5 hours

Problem 7 (12RMM1)

We create a bipartite graph where the boys and girls are vertices and edges are drawn between a pair of different genders that know each other. We strong induct on how many edges there are.

Base case: 0 edges. Trivially, either there is one way on each side or zero ways on each side (if there are at least one boy, one girl).

Inductive step:

We select one of the edges to remove between boy A and girl B to make G' . All sets of sociable boys and sociable girls that satisfy G' are still sociable in G . By the inductive hypothesis, there are the same parity of these sets. Therefore, to finish, we show that the number of sets of boys that have changed to become *sociable* with the addition of the edge between A, B has the same parity as changed sets of sociable girls.

For a set of boys, for example, to become sociable with the addition of the edge, note that the set must include A such that adding an edge does something. The set must also have previously been disconnected from B . Therefore, the set must not include any of the boys apart from A that B has connections with. The condition to satisfy is for the other boys in the set must reach all the girls apart from those that A reach. Therefore, this is the same as the number of *sociable set of boys* on a graph G'' where we have removed A, B (and the edge between them), and all the vertices (and edges connected to those vertices) that are adjacent to A or B .

Symmetrically, the number set of girls that have become sociable is the number of *sociable sets of girls* on G'' . By the inductive hypothesis, since there are less edges in G'' , we are done.

Conclusion: By induction we are done.

Problem 8 (02IMO1)

(This solution may have some typos but I think you get the point)

Let $r_0, r_1, \dots, r_n$ be the number of blue cells in each row from **top to bottom**. We have $B = r_0 r_1 \dots r_n$.

Let $c_0, c_1, \dots, c_n$ be the number of blue cells in each column from **left to right**. We have $A = c_0 c_1 \dots c_n$.

We prove that $\{r_0, r_1, \dots, r_n\}$ is a permutation of $\{c_0, c_1, \dots, c_n\}$, hence proving $A = B$.

We prove this using induction on n .

Base case: $n = 1$. There are two colorings so trivial.

Inductive step:

Let the bottom row have k reds filling the leftmost k lattice points. No other rows may currently have more than k reds, so $c_k = n - k + 1, c_{k+1} = n - k, \dots, c_n = 1$. We define for now, $c_{n+1} = r_{n+1} = 0$.

We add a new row of $n + 1$ points below this row. Let there be $j \geq k$ reds on this row filling the leftmost k lattice points. Note that, $c_0, \dots, c_{j-1}$ do not change, nor does $r_0, \dots, r_n$. However, $c_j, \dots, c_{n+1}$ are all increased by one (recall c_{n+1} was previously zero) and $r_{n+1} = 0$ is replaced with $r_{n+1} = n - j + 1$.

Since $c_j, \dots, c_{n+1}$ were always consecutive decreasing, increasing them all by one is the same as replacing the minimum with one more than the maximum. Indeed, this is what we did to r .

Therefore, after the move, $\{r_0, r_1, \dots, r_{n+1}\}$ is also a permutation of $\{c_0, c_1, \dots, c_{n+1}\}$.

Conclusion: By mathematical induction we are done.

14 clubs; 6 hours

Problem 10 (17SLC3)

Note that the arrangement of the cells does not matter. Let $f(n, m)$ be maximum number of ways to generate 2^n with m cells. Also note that the sum of the numbers in the board is non-strictly increasing so we may never choose a number greater than 2^n . Therefore, note that $f(n, 1) = f(0, m) = 1$ always with only one possible move.

For $n > 0, m > 1$, note that unless if we want to spawn in 2^n first move and immediately end the game, our last move would be to combine two 2^{n-1} s. Note that, clearly, $f(n, m) \leq f(n, m + 1)$ so we want the most space to create each of the 2^{n-1} s to maximize moves. Therefore, to maximize moves, we first create one of the 2^{n-1} s using the m spaces, then create the second using the $m - 1$ spaces [clarification needed on why this maximizes?]. Hence, in this case,

$$f(n, m) = f(n - 1, m) + f(n - 1, m - 1) + 1$$

We denote $g(n, m) := \frac{f(n, m) + 1}{2}$. Note that, we still have, $g(n, 1) = g(0, m) = 1$. Also,

$$f(n, m) = f(n - 1, m) + f(n - 1, m - 1) + 1$$

$$\Leftrightarrow g(n, m) = g(n - 1, m) + g(n - 1, m - 1)$$

Instead, we define a function $g'(n, m)$ where all conditions are same, except, we define $a_m := g(0, m)$.

By induction, we may find that $g'(n, m) = \binom{n}{0} a_m + \binom{n}{1} a_{m-1} + \dots + \binom{n}{m} a_0$. Plugging in $a_0 = 0; a_1 = a_2 = \dots = a_m = 1$, we have $g(n, m) = \binom{n}{0} + \dots + \binom{n}{m-1}$. Therefore,

$$f(n, m) = 2 \left(\binom{n}{0} + \dots + \binom{n}{m-1} \right) - 1$$

Problem 11 (15RUS113)

We call the condition that out of every group of 55 teams, there exists a team which has lost to at most four of the other 54 teams in the group the “fifty-five condition”. We induct on the number of teams N .

Base case: $N = 55$. Trivially true by the “fifty-five condition”.

Inductive step: $N \rightarrow N + 1$

For the sake of contradiction, let there be a way $N + 1$ teams played the tournament such that each team lost to at least 5 of the other N teams given the “fifty-five condition”. We claim that there is a way to remove a team from the game such that the remaining teams still satisfy each team losing to at least 5 of the other N teams.

Let x be the number of teams which have exactly 5 losses. With $N + 1$ teams, the sum of the number of wins over all teams is $\frac{N(N+1)}{2}$. Therefore, each team with exactly 5 losses has $N - 5$ wins for a total of $x(N - 5)$ wins. The remaining $N + 1 - x$ teams have $\frac{(N-x)(N+1-x)}{2}$ wins in the games they play amongst themselves.

$$\begin{aligned}
& x(N-5) + \frac{(N-x)(N+1-x)}{2} \leq \frac{N(N+1)}{2} \\
\Leftrightarrow & x(N-5) + \frac{N(N+1) - x(2N+1) + x^2}{2} \leq \frac{N(N+1)}{2} \\
\Leftrightarrow & x(N-5) + \frac{-x(2N+1) + x^2}{2} \leq 0 \\
\Leftrightarrow & -5x + \frac{-x + x^2}{2} \leq 0 \\
\Leftrightarrow & 0 \leq x \leq 11
\end{aligned}$$

We may simply avoid the at most 55 teams at least one of these x teams lost against to remove. Since we had $N+1 \geq 56$ teams to choose from, this is possible. Removing this team, the number of losses each team has is still at least 5 as desired.

Since removing this team will not make the “fifty-five condition” untrue, we’d have N teams that contradict the inductive hypothesis.

Conclusion: By induction $N = 110$ works.

22 clubs; 8.5 hours

Problem 12 (18USATST4)

We prove this by induction on n .

Base case: $n = 1, 2$. P_k is true for all k for $n = 1$ and $n = 2$ may be checked by hand.

Inductive step:

Note that, if we add a bit to the i th position of each element of S to get the length from n to $n+1$, P_k cannot go from false to true since the unchanged n digits of all the strings will remain the same. P_k goes from true to false only if there were $2k+1$ binary strings $x_1, x_2, \dots, x_{2k+1}$ whose majority x used to be an element $x = m \in S$. However, their majority is no longer in S . Let m' be m after receiving the new digit and define $x'_1, \dots, x'_{2k+1}, x'$ analogously. We have $x = m$ previously but $x' \neq m'$ now. Therefore, the last digits of x', m' must be distinct causing P_k to no longer be true.

Therefore, if we flip a bit of x' the majority of $x'_1, \dots, x'_{2k+1}$ at any position i to get x_i^* we have the result is in S . For any other P_j , we take $x_1^*, x_2^*, \dots, x_{2j+1}^*$ (indices modulo $2k+1$) to get x' as the majority, meaning P_j fails as well.

Conclusion: Starting with $n = 1$, P_k is initially true for all n . However, adding digits, when there is a point where P_k becomes false for some k , all P_k go to false. This finishes the problem.

Problem 14 (CF1586F)

The answer is $\lceil \log_k n \rceil$.

Construction: <https://codeforces.com/contest/1586/submission/267913041>

Proof: Problem 5 (15USATST). Instead of splitting into A (starts) and B (ends) we split into A_0 (starts), A_1 (at most edge 1 from a start), ..., A_{k-1} (ends).

32 clubs; 11 hours

Problem 16 (09USATST1)

We prove this using induction on m, n . Let $f(m, n)$ be the number of ways to fill the rectangle choosing m, n .

Base cases: $\min(m, n) = 1$

WLOG let $m = 1$. With a corner removed, we are filling a $1 \times 2^n - 1$ rectangle with $1 \times 2^0, 1 \times 2^1, \dots, 1 \times 2^n$ tiles. There are evidently $n! < (n + m)!$ permutations ways to do this.

Inductive step:

To fill the $2^m \times 2^n$ rectangle with a corner removed, we look at how to use the rectangle with area 2^{m+n-1} . Note that since the side lengths are integers and the area is a power of two, the side lengths must also be a power of two. There are only two options which may fit into the $2^m \times 2^n$ rectangle which are $2^{m-1} \times 2^n$ or $2^m \times 2^{n-1}$.

We first look at $2^{m-1} \times 2^n$. Placing this would divide the remainder of R into a $x \times 2^n$ rectangle with a corner removed and a $(2^{m-1} - x) \times 2^n$ rectangle for some $1 \leq x \leq 2^{m-1}$. If we combine the remaining pieces, we have a $2^{m-1} \times 2^n$ rectangle with a corner removed. Tiling in the remaining tiles of area $2^0, 2^1, \dots, 2^{m+n-2}$ we can draw a vertical line at the x -th column not cutting through any tile. Therefore, for a tiling T of a $2^{m-1} \times 2^n$ rectangle with a corner removed, if $f(T)$ is the number of vertical lines through T not cutting through any tiles including the right edge but not the left (since inserting the $2^{m-1} \times 2^n$ rectangle at the left would cause the removed corner to no longer be in a corner). The number of cases here is $\sum_T f(T)$. [clarification needed?] We show that $\sum_T f(T) \leq m(m + n - 1)!$.

By the inductive hypothesis, there were $(m + n - 1)!$ tiling of this rectangle. We show that $f(T) \leq m$ for each tiling T . For the tile with area 2^{m+n-2} , if we place it as a $2^{m-1} \times 2^{n-1}$ tile, then $f(T) = 1$. Therefore, to maximize $f(T)$, we must place it vertically. Then, same thing for the tile of areas $2^{m+n-3}, \dots, 2^n$. The remaining tiles would then be cramped into one column. In total, $f(T) = m$ at max and we are done.

Analogously, if we used the $2^m \times 2^{n-1}$ piece, there are at most $n(m + n - 1)!$ cases. In total, that's at most $(m + n)(m + n - 1)! = (m + n)!$ in total.

Problem 19 (04GERMANTST)

We induct on the number of vertices.

Base case: $n = 1$. We flip the only vertex.

Inductive step: $n \rightarrow n + 1$

For the sake of contradiction, if the problem is possible for any graph with n vertices but not a graph G with $n + 1$ vertices, we there are sequences of moves $S_1, S_2, \dots, S_{n+1}$ where S_i flips everything in the graph except for vertex i . If n is odd, we may do sequences $S_1 + S_2 + \dots + S_{n+1}$ to flip all the vertices n times and finish.

If n is even, note that if we do $S_i + S_j$, we may flip i, j to red and everything else stay as black. Therefore, we may flip the colors of any subset of G with an even number of elements. If there is any element of G with an even degree, we flip that vertex to flip an odd number of vertices. Since G has an odd number of vertices in total, there is a way to flip the remaining vertices. If all the elements of G have an odd degree, then since $|G|$ is odd the sum of the degrees of all the vertices of G is odd, which is a contradiction as each edge contributes 2 to the total degree.

42 clubs; 14 hours

Problem 21 (17RMM3)

The answer is $2n - 2$. Label the elements of X to be $X = \{1, 2, \dots, n\}$.

To prove attainability, we take $\{1\}, \{2\}, \dots, \{n - 1\}$ and $X - \{1\}, X - \{2\}, \dots, X - \{n - 1\}$. This works as choosing a collection from only the first $n - 1$ subsets of magnitude 1 cannot satisfy no element of X lies in exactly one of the A_i 's, meaning we must choose from one of the $X - \{i\}$'s. However, choosing $X - \{i\}$ introduces n , meaning we must choose another $X - \{j\}$ such that n lies in at least two of the subsets. However, by choosing that, the union is X .

Before we prove the bound, we cover the following lemmas.

Lemma 1: A non-empty collection of subsets $A_1, \dots, A_k$ of X is *rigid* if no element of X lies in exactly one of the A_i 's. The largest cardinality of a collection of non-empty subsets of $\{1, \dots, n\}$ such that no non-empty subcollection of which is *rigid* is n .

Proof: To attain n we choose $\{1\}, \{2\}, \dots, \{n\}$. To prove the bound we use induction.

Base case: $n = 1$. The only non-empty subset is $\{1\}$. Choosing $\{\{1\}\}$ we are done.

Inductive step:

For the sake of contradiction, let the answer to $\{1, \dots, n\}$ be $n + 1$. Note that for our choice of $A_1, \dots, A_{n+1}$, if there exists some $i \in \{1, \dots, n\}$ that is not present in any subset, then $A_1, \dots, A_{n+1}$ are subsets of $\{1, \dots, n\} - \{i\}$ which contradicts the inductive hypothesis of there being at most $n - 1$. If all $i \in \{1, \dots, n\}$ are present in at least two of $A_1, \dots, A_{n+1}$, then choosing the entire collection $A_1, \dots, A_{n+1}$ is *rigid* (!). Therefore, there must be some $i \in \{1, \dots, n\}$ present in exactly one of $A_1, \dots, A_{n+1}$. Let's say i is present in A_{n+1} WLOG. The sets $A_1, \dots, A_n$ are subsets of $\{1, \dots, n\} - \{i\}$ which once again contradicts the inductive hypothesis of there being at most $n - 1$.

Conclusion: The bound is proven and the lemma is proven.

We now finish the problem.

Base case: $n = 2$. We have $\{1\}, \{2\}$ are the only non-empty proper subsets. Using both clearly works.

Inductive step:

For the sake of contradiction, if it were possible to do $2n - 1$, then for any $i \in X$, there are either at least n sets without i or n sets with i . If there were n sets without i , we would have a collection C of n subsets of $\{1, \dots, n - 1\}$ for which all subcollections are not *rigid*, which contradicts the lemma (!).

Therefore, there must be n sets with i for all i .

We claim that $\{1\}, \{2\}, \dots, \{n\}$ all must be chosen in this case. FTSOC, if this was not the case and WLOG $\{n\}$ wasn't present, then let $A_1, \dots, A_n$ be the n sets from our collection which have n as an element. If we take n out of each of these sets, the resulting $A'_1, A'_2, \dots, A'_n$ are non-empty proper subsets of $\{1, \dots, n - 1\}$. For subcollections of $A_1, \dots, A_n$ to be not *tight*, subcollections of $A'_1, \dots, A'_n$ must be not *rigid* which once again contradicts the lemma as desired.

Therefore, with $\{1\}, \dots, \{n\}$ each being chosen, since $2n - 1 > n$ for $n \geq 2$, there must be another proper subset S of X in our collection as well with $|S| > 1$. If $S = \{a_1, \dots, a_m\}$, we may choose $S, \{a_1\}, \{a_2\}, \dots, \{a_m\}$ to form a *tight* collection of subsets of X . (!)

Conclusion: By induction we're done.

Problem 22 (21TSTST7)

We induct on n .

Base case: $n = 1$. Can be checked to be true.

Inductive step:

We casework on the first step.

Case 1: Both $(1, 0)$ and $(0, 1)$ have ways out.

By the inductive hypothesis, there are at least $2^{n-1-|M|}$ ways each.

Case 2: Both $(1, 0)$ and $(0, 1)$ have no way out.

There is then no way out from $(0, 0)$, which is acceptable.

Case 3: One of $(1, 0)$ or $(0, 1)$ has no way out.

WLOG, let there be no way out from $(1, 0)$ but a way out from $(0, 1)$. One of the mines must be on at least one of $(1, 0), (2, 0), \dots, (n, 0)$ meaning there are at most $|M| - 1$ remaining mines which can affect $(0, 1)$. There are then $2^{n-1-(|M|-1)} = 2^{n-|M|}$ paths from $(1, 0)$ as desired.

56 clubs; 17 hours